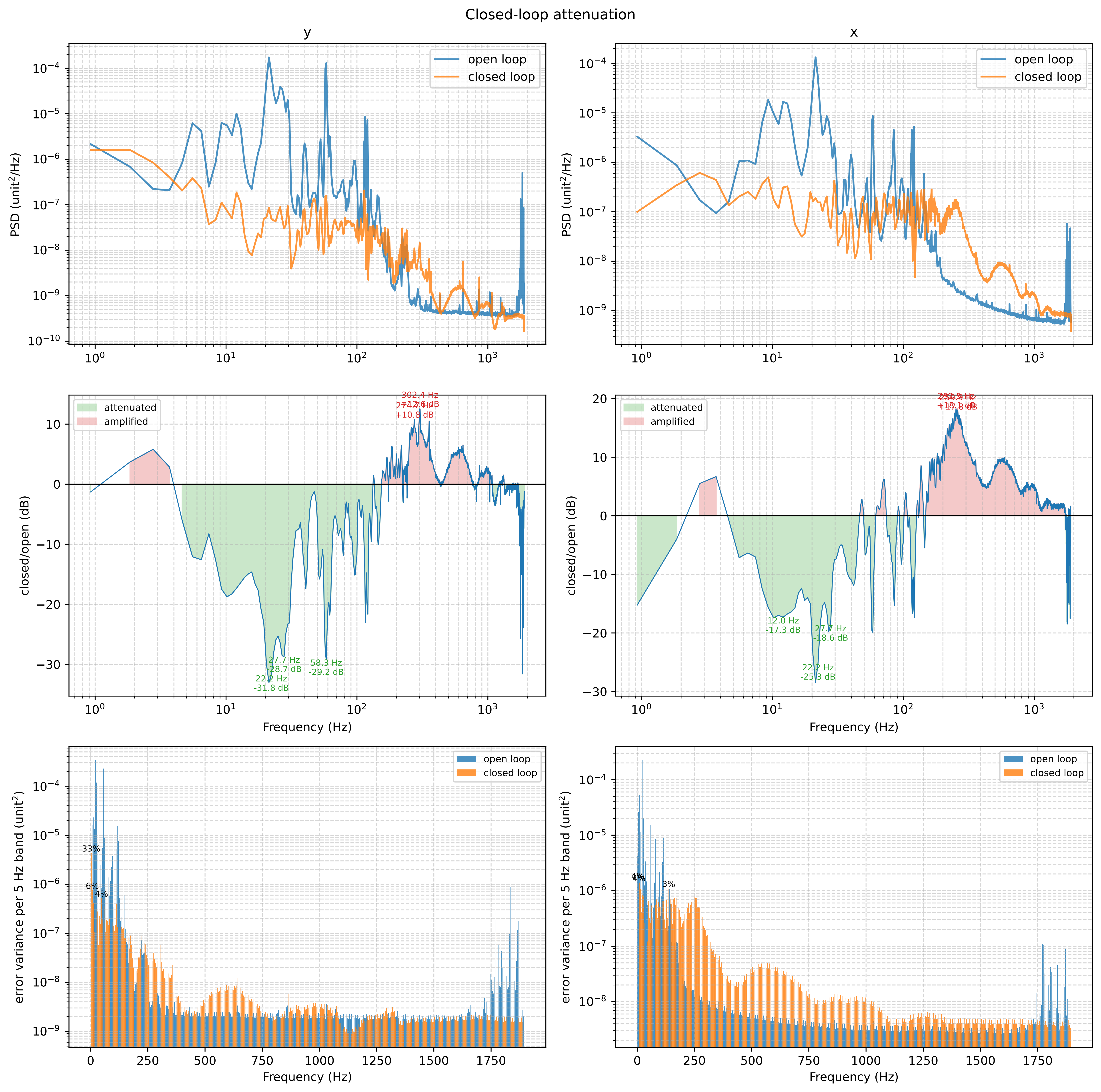
Closed-loop attenuation
y
x
PSD (unit^2/Hz)
open loop
closed loop
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{-9}
10^{-10}
10^0
10^1
10^2
10^3
attenuated
amplified
closed/open (dB)
302.4 Hz
+12.6 dB
274.7 Hz
+10.8 dB
27.7 Hz
-28.7 dB
22.2 Hz
-31.8 dB
58.3 Hz
-29.2 dB
252.5 Hz
+18.1 dB
259.9 Hz
+17.6 dB
12.0 Hz
-17.3 dB
27.7 Hz
-18.6 dB
22.2 Hz
-25.3 dB
20
10
0
−10
−20
−30
Frequency (Hz)
error variance per 5 Hz band (unit^2)
33%
6%
4%
4%
4%
3%
0
250
500
750
1000
1250
1500
1750

Run configuration and per-band RMS

```
Started: 2026-07-31 18:33:23
pixel_scale: 15.5 px per output unit
y: mean -0.482 -> -0.001 px;  RMS about mean 0.511 -> 0.057 px   (open -> closed)
x: mean +3.540 -> -0.000 px;  RMS about mean 0.378 -> 0.094 px   (open -> closed)

Config used:
ATTENUATION PAR-6 -- PAR-5 REPEATED with broad_lp 5 -> 9 as the ONLY change anywhere in the file (see _broad_lp for the measurement that motivates it). Timeline, ROI, K, delay, mu schedule, guard and modal table are all byte-identical to PAR-5, so PAR-5 is the direct comparison and y is the CONTROL: x should improve substantially above 100 Hz and y should not move more than a few percent. Everything from here down describes PAR-5 and still applies. PAR-5 was: PAR-4 REPEATED with the beam RE-CENTRED (start_y 684 -> 692, park verified +0.70 px y / +0.53 px x, against PAR-4's +5.95 px which ran the 20x20 tracking window off the ROI edge and halved y's rejection in every band) and with the NLMS step SCHEDULED 0.5 -> 0.03 over 60 s to fix the convergence PAR-4's trace showed was still incomplete. Kx stays 4.05 -- PAR-4 proved it: x improved in every band (60-150 Hz 1.4x -> 2.9x, 150-250 Hz from 0.4x amplifying to 1.3x rejecting) and beat its environment-scaled predicted asymptote by 11%. Timeline unchanged from PAR-4 so the scored windows are directly comparable. Through the PARPORT DAC (TI DAC81404 over the StarTech PEX1P2 / AX99100 PCIe parallel card, spi link, mmio backend), replacing the DAQC2/BRIDGEplate serial path used by every previous attenuation run.
TRANSPORT: bit-banged 24-bit SPI, ~6.4 us per channel update (~12.8 us for both channels), against the BRIDGEplate's ~0.5-1.4 ms. DAC range is FIXED at +-5 V bipolar (153 uV/LSB) with the mirror parked at 0 V, matching the bode sweeps exactly; auto-ranging is off, so the actuator is a fixed-gain element for both phases of the A/B and the open/closed comparison is clean. Supplies +/-12 V: AVDD needs VMAX+1.5 = 11.5 V for +-10, AVSS -11.5 V, leaving only 0.5 V of margin -- meter the rails under load. AVSS is therefore NEGATIVE, not grounded, which means the 0-5 base range runs outside the AVSS = 0 V condition SLASEH2A 7.5 gives the unipolar figures at (see _avss_tradeoff); 16-bit = 76 uV/LSB in 0-5, 305 uV/LSB in +-10, vs the BRIDGEplate's 12-bit 1 mV.
OUTPUTS: two channels, no coarse stage -- ch0 = y/tip (+1.0 V/unit), ch1 = x/tilt (-1.0 V/unit), offset 0 V, symmetric command bounds +-1 V (59 px of authority against a +-16 px ROI; the mirror's full range is deliberately NOT given to the loop on this first run). NOTE the command->FSM voltage mapping is NOT the same as any previous attenuation run, so K is not comparable across the transport change and was re-measured -- and it came out 4.7-5.5x HIGHER, not lower as this file originally predicted, so nothing about the piplate-era gains carries over.
Camera/CoM: 32x32 ROI start_x 1832 / start_y 662 (re-derived 2026-07-31 immediately before PAR-6; park measured 4x at exposures 50/50/20/10 reading x 1846.2/1845.7/1846.2/1846.0 and y 677.2/677.1/677.2/676.8, std 0.24/0.19 px, giving a beam offset of -1.3 px x / -0.3 px y from frame centre and 4.7 px of tracking-window clearance -- against the 1.0 px clearance that a +5 px park had left earlier the same day, and PAR-4's edge-clipping park), exposure 50, gain 100, bw 100, 20x20 tracked window thr 5 (the camera is now shrouded, so the background floor no longer needs a high threshold). Identical to conf_bode_par_x/y[_small].json, so K and tau were fit through exactly this optical chain.
FSP: broad_order 512, broad_mu .03 -- REVERTED from PAR-2's 128/.005, which was 60% worse on y and 10% on x once normalised for the environment. 128 taps = 29.6 Hz resolution cannot separate the 2-3 Hz-spaced 9-30 Hz comb; this reproduces the documented kalman order-80 failure (order 80 ~29 Hz could not represent the 2.256 Hz comb). broad_lp 9 in PAR-6, 5 in PAR-1..PAR-5 (see _broad_lp: 5 taps do cover the 685 Hz delay ring they were chosen for, but give only -2.0 dB at the 280 Hz command-noise hump that carries 90% of x's closed variance); adapt_period 8, adapt_tau 5. Guard ON (guard_ratio 4). Cold-start observer: pass_open false + broad_freeze_closed false, so no weight is derived from open-loop data; drift_tau and the transient path are OFF. Modal freqs re-derived from run 1's own open PSD (8-mode cap), though they are inert while broad_order > 0.
PURPOSE: isolate the two things PAR-4 confounded. PAR-4 changed Kx AND happened to run with the beam 5.95 px off centre; x improved everywhere (the Kx fix) while y lost half its rejection uniformly across 1-60 Hz (the clipped tracking window). PAR-5 fixes the centring and changes nothing else about the controller except the step schedule, which the offline replay shows does not move the converged solution. EXPECTATIONS: y should return to a closed/open ratio near PAR-3's 0.121 (PAR-4 was 0.233); x should hold or beat PAR-4's 0.248 (PAR-3 was 0.302). Prior absolute numbers are NOT directly comparable because the environment moves run to run -- PAR-4's open was 18% (y) and 34% (x) larger than PAR-3's, and a new 150-250 Hz component appeared that carried 8.7x (y) and 5x (x) more open power than in PAR-3. Always normalise the closed RMS by that run's own open RMS before claiming an improvement.
DELAY AND K: taken from different sweeps on purpose. DELAY from the newest full 5-300 Hz sweeps, fsm_bodex_par2 / fsm_bodey_par2 (2026-07-31 12:23 / 12:16): x 2.763 fr (tau 0.4654 ms), y 2.624 fr (tau 0.4288 ms) at fs 3788. PAR-1..3 ran 2.717 / 2.577; the +0.046 fr shift sits inside the fits' own phase residuals, so the two sweep pairs agree and the newer/longer value is kept only because under-modeled delay is the documented failure mode. K from the SMALL-SIGNAL plateau: Kx 4.05, Ky 3.646. That distinction is not academic -- the loop commands ~6 mV rms / ~20 mV peak and the FSM compresses ~4% at +-100 mV, but WHICH AXIS compresses moves between days: on 07-30 it was x (100/50 mV ratio 1.041, y flat at 0.998), on 07-31 it was y (ratio 1.039, x flat at 1.003). The raw 07-31 fit lines therefore read Kx 4.050 (correct) and Ky 3.510 (3.7% low), and PAR-4 uses 3.646 for y from the +-20 mV sweep, corroborated by the same full file's +-50 mV pass at 3.662. Conversely tau is NOT measurable from a small sweep -- 10-30 Hz spans ~7 deg of phase and their loop_frames scatters 1.09-2.02 fr -- which is why delay comes only from the full sweeps. Against the piplate era this is ~5x MORE loop gain and LESS THAN HALF the delay; the shorter delay is the entire reason to expect better rejection here. Caveat: the small-signal gain itself wanders a few percent per hour (seven +-20 mV x sweeps read 3.97-4.36, each internally consistent), and erring HIGH on K is the safe direction since u = -phi_hat/K.
COMPARE against attenuation_fsp21 and attenuation_fsp15 (run 15: y 0.073 / x 0.168 px): report total and per-band closed RMS, 60-400 Hz waterbed, command/clamp behavior, and transient activation count on the shared 500/500/4096 grid. Closed RMS in PIXELS remains comparable across the change; command units do not. refs: Kulcsar 2006 / Petit 2008 / Meimon 2010 / Correia 2010
```

RMS contributed by each 10 Hz band (px)

band (Hz)	y open	y closed	x open	x closed
0-10	0.0696	0.0348	0.0848	0.0264
10-20	0.0933	0.0112	0.1239	0.0188
20-30	0.3317	0.0120	0.2429	0.0199
30-40	0.0505	0.0082	0.0332	0.0125
40-50	0.0280	0.0127	0.0220	0.0137
50-60	0.2367	0.0115	0.0627	0.0119
60-70	0.0480	0.0084	0.0132	0.0144
70-80	0.0241	0.0088	0.0212	0.0187
80-90	0.0204	0.0101	0.0533	0.0158
90-100	0.0377	0.0086	0.0265	0.0165
100-110	0.0140	0.0079	0.0165	0.0167
110-120	0.0704	0.0112	0.0541	0.0151
120-130	0.0445	0.0073	0.0382	0.0146
130-140	0.0097	0.0079	0.0100	0.0184
140-150	0.0164	0.0071	0.0199	0.0175
150-160	0.0068	0.0066	0.0075	0.0152
160-170	0.0045	0.0056	0.0069	0.0175
170-180	0.0052	0.0060	0.0075	0.0171
180-190	0.0021	0.0035	0.0048	0.0138
190-200	0.0020	0.0026	0.0038	0.0107
200-210	0.0024	0.0035	0.0033	0.0126
210-220	0.0028	0.0044	0.0029	0.0138
220-230	0.0051	0.0056	0.0029	0.0156
230-240	0.0043	0.0054	0.0028	0.0168
240-250	0.0032	0.0040	0.0028	0.0181
250-260	0.0014	0.0034	0.0025	0.0188
260-270	0.0012	0.0035	0.0023	0.0156
270-280	0.0013	0.0041	0.0024	0.0142
280-290	0.0014	0.0033	0.0023	0.0116
290-300	0.0017	0.0043	0.0023	0.0099
total 0-300	0.4420	0.0532	0.3109	0.0880